

Math 142, F19
Quiz 11

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (10 points) Find the Maclaurin series and its radius of convergence for the following functions.

(a) $f(x) = xe^x$

$$e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \text{ for } |x| < +\infty$$

$$\Rightarrow f(x) = xe^x = x \sum_{n=0}^{+\infty} \frac{x^n}{n!} = \sum_{n=0}^{+\infty} \frac{x^{n+1}}{n!}, \text{ (or } \sum_{n=1}^{+\infty} \frac{x^n}{(n-1)!})$$

$$|x| < +\infty \quad (\Rightarrow R = +\infty)$$

(b) $f(x) = \sin(2x)$

$$\sin x = \sum_{n=0}^{+\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, \text{ for } |x| < +\infty$$

$$\Rightarrow \sin(2x) = \sum_{n=0}^{+\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=0}^{+\infty} \frac{(-1)^n 2^{2n+1} x^{2n+1}}{(2n+1)!}, \text{ for } |2x| < +\infty$$

$$\Leftrightarrow |x| < +\infty \quad (\Rightarrow R = +\infty)$$

(c) $f(x) = (1-x)^{-2}$

(Hint: $(1+x)^k = \sum_{n=0}^{+\infty} \frac{k(k-1)\dots(k-n+1)}{n!} x^n$)

$$(1-x)^{-2} = [1+(-x)]^{-2} = 1 + \frac{(-2)}{1!}(-x) + \frac{(-2)(-3)}{2!}(-x)^2 + \frac{(-2)(-3)(-4)}{3!}(-x)^3 + \dots$$

$$+ \frac{(-2)(-3)\dots(-1-n)}{n!}(-x)^n + \dots$$

$$= 1 + \frac{2}{1}x + \frac{2 \cdot 3}{2!}x^2 + \frac{2 \cdot 3 \cdot 4}{3!}x^3 + \dots + \frac{2 \cdot 3 \cdot 4 \dots (n+1)}{n!}x^n + \dots$$

$$= \sum_{n=0}^{+\infty} (n+1)x^n, \text{ for } |x| < 1 \quad (\Rightarrow R=1)$$

$$\Rightarrow |x| < 1$$